

Instrument Approach Procedure Chart Georeferencing: PDF Boxes

[ACM Charting Group 26-01-411](#)

2026-08-21

Mark Mentovai
mark@mentovai.com

00610IL22L.PDF

NEW YORK, NEW YORK

AL 51.1 (FAA) **594**

25275

ILS or LOC RWY 22L
JOHN F KENNEDY INTL (JFK)

LOC/DME HWY 110.9
Chan 46

APP CRS 224°
TDZE 12
Rwy Ldg 8400
TDEZ 12
Apt Elev 13

RNAV 1-GPS or RADAR required for procedure entry.
From CIMBL, NRTON: RNAV 1-GPS required.

Simultaneous approach authorized.

ALSIF-2

MISSED APPROACH: Climb to 500 then climbing left turn to 3000 on JFK VOR/DME R-190 to CHANT INT/19 DME and hold.

(SW) (NE)
D-ATIS (NW) (SE)
115.4 117.7 128.725
NEW YORK APP CON 128.125 269.0
KENNEDY TOWER 119.1 281.55 (4R/22L and 13U/31R)
123.9 281.55 (4L/22R and 13R/31L)
GND CON 121.9 348.6

← **MediaBox, CropBox**

← **0**

→ **387.36**

NE-2, 06 AUG 2026 → 03 SEP 2026

NE-2, 06 AUG 2026 → 03 SEP 2026

NEW YORK, NEW YORK
Amdt 26 31DEC20

40°38'N 73°47'W

ILS or LOC RWY 22L
JOHN F KENNEDY INTL (JFK)

RLS Rwy's 13L and 13R
HIRL all Rwy's
TDZ/CL Rwy's 4L, 4R, 13L, 22L and 31R
FAF to MAP 5.5 NM
Knots 60 90 120 150 180
Min/Sec 5:30 3:40 2:45 2:12 1:50

The chart depicts the Instrument Landing System (ILS) and Localizer (LOC) for Runway 22L at John F. Kennedy International Airport. Key features include:
 - **Navigation Aids**: VORTAC station JFK (110.9 MHz, Chan 46), VORTAC HAUTP (113.9 MHz, Chan 106), and VORTAC SAJUL (113.9 MHz, Chan 106).
 - **Communication**: Tower frequency 119.1 MHz, Ground Control 121.9 MHz, Unicom 123.9 MHz, ATIS 115.4 MHz, and Enroute frequencies 128.125 MHz and 269.0 MHz.
 - **Procedures**: Standard ILS and LOC approaches are shown with altitude restrictions (e.g., 3000 ft, 1800 ft, 1500 ft) and speed limits (e.g., 200 kts, 150 kts).
 - **Missed Approach**: Instructions for missed approach are provided, including a climb to 500 ft followed by a left turn to 3000 ft on the JFK VOR/DME R-190 radial to Chant Int/19 DME and hold.
 - **Obstacles**: Obstacle symbols are present near the runway end, including one at 3000 ft MSL and 1500 ft AGL.
 - **Charts and Insets**: Various inset charts show nearby airports (e.g., Westbury, Hempstead Harbor, Jamaica Bay) and provide additional navigation information.
 - **Annotations**: Large blue arrows point to the runway number 594 and the runway name. A red box highlights the MediaBox and CropBox areas. A red circle with the number 0 is also visible.

← MediaBox, CropBox

115.9 JFK Chan 106 10000 to C Chan 46 SAJULI (not to st) **387.36**

[illegible]

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Page→VP[0]→BBox

This defines the georeferenced viewport.

Viewport: “a rectangular region of the page”.

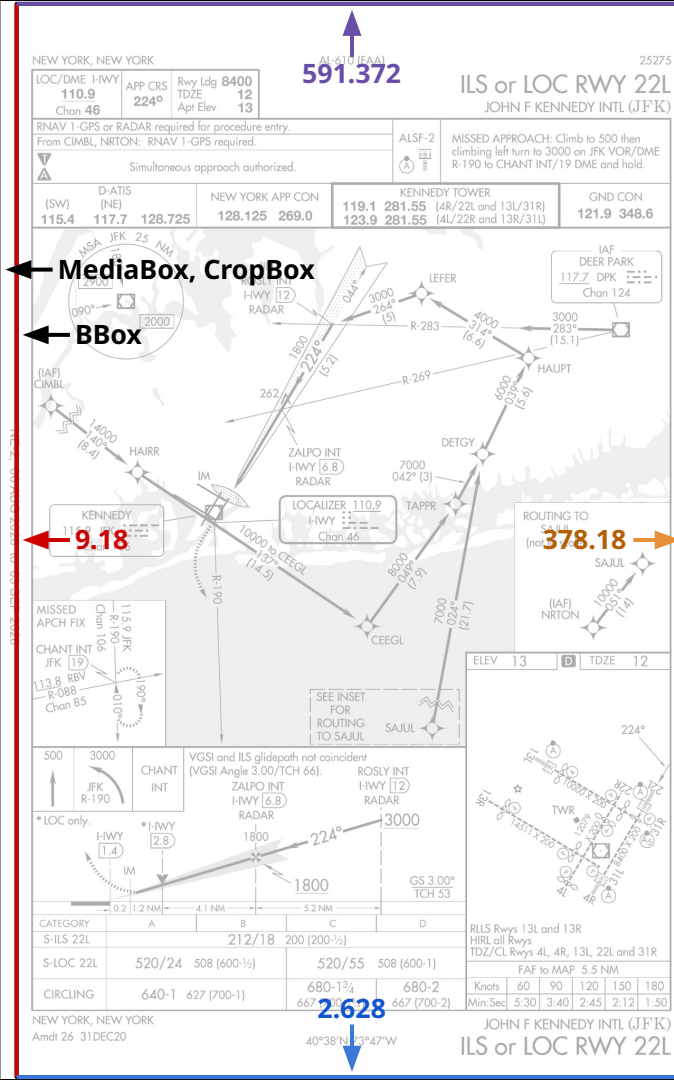
BBox: “a rectangle [...] specifying the location of the viewport on the page”.

Like MediaBox and CropBox, this must be a rectangle.

It is incidentally very slightly smaller than the MediaBox and CropBox:
.95 width, .99 height, and .94 area.



00610IL22L.PDF



Page→VP[0]→Measure→Bounds



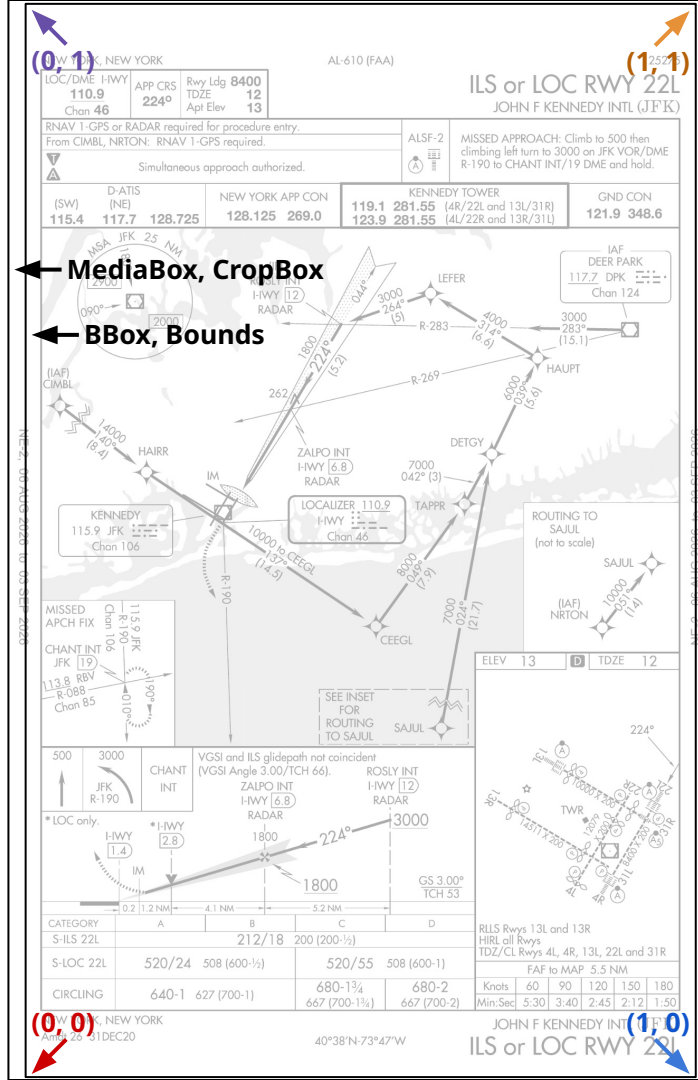
Taken relative to the BBox, this defines the neatline.

Bounds: “a series of points that describes the bounds of an area for which geospatial transformations are valid. [...] The numbers are expressed relative to [...] the BBox associated with a Viewport [...]”.

00610IL22L.PDF

Bounds can be a polygon, it doesn't need to be rectangular, although it is here.

While it would be best to match the boundaries of the plan view on the chart, it currently matches the full extent of the BBox: .94 of the page (MediaBox/CropBox) area.



00610IL22L.PDF

[illegible]

← **MediaBox, CropBox**

← BBox, Bounds

CIMBL ← LPTS

NEW YORK, NEW YORK
Amdt 26 31DEC20
40°38'N-73°47'W
JOHN F KENNEDY INTL (JFK)
ILS or LOC RWY 22L

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00610IL22L.PDF

N 41°04'50.83"
W 74°01'12.98"

(-0.024, -1.004)

AL-610 (FAA)

(1.024, 1.004)

N 41°04'48.76"
W 73°12'23.08"

JOHN F KENNEDY INTL (JFK)
MISSED APPROACH: Climb to 3000
climbing left turn to 3000 on JFK VOR/DME
R-190 to CHANT INT/19 DME and hold.

Simultaneous approach authorized.

(SW)	D-ATIS	(NE)	NEW YORK APP CON	KENNEDY TOWER	GND CON
115.4	117.7	128.725	128.125 269.0	119.1 281.55 (4R/22L and 13L/31R) 123.9 281.55 (4L/22R and 13R/31L)	121.9 348.6

MediaBox, CropBox, LPTS

BBox, Bounds

MISSED APCH FIX
CHAN INT JFK [19]
113.8 RBV
R-088
Chan 85

ROUTING TO SAJUL
(not to scale)

SEE INSET
FOR
ROUTING TO
SAJUL

CATEGORY	A	B	C	D
S-ILS 221	212/18	200 (200-½)		
500/55	508 (600-1)			
680-1%	680-2			
667 (700-1)	667 (700-2)			

N 40°08'11.55"
W 74°00'56.06"

(-0.024, -0.004)

N 40°08'09.52"
W 73°12'48.26"

(1.024, -0.004)

ILS or LOC RWY 22L

LPTS: “[...] define points **in** a 2D unit square [...] mapped to the rectangular bounds of the viewport [...] each number pair is the unit square object position corresponding to the geospatial position in the GPTS array”.

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endobj
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00610IL22L.PDF

(0, 1)

NEW YORK
LOC/DME RWY 13L

N 41°04'50.83"
W 74°01'12.98"

AL-610 (FAA)

(1, 1)

ILS or LOC RWY 22L
JOHN F KENNEDY INTL (JFK)
W 73°12'23.08"

Simultaneous approach authorized.

(SW)		(NE)		NEW YORK APP CON		KENNEDY TOWER		GND CON	
115.4	117.7	128.725	128.125	269.0	119.1	281.55	(4R/22L and 13L/31R)	121.9	348.6
					123.9	281.55	(4L/22R and 13R/31L)		

MediaBox, CropBox, BBox, Bounds, LPTS

CHART DATA:

CATEGORY	A	B	C	D
S-ILS 22L	212/18	200 (200-½)		
5-ILS 22L	520/55	508 (600-1)		
680-1%	680-2%	680-2%		
667 (700-1%)	667 (700-2%)			

(0, 0)

NEW YORK
LOC/DME RWY 13L

N 40°08'11.55"
W 74°00'56.06"

JOHN F KENNEDY INTL (JFK)
W 73°12'23.08"

ILS or LOC RWY 22L

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<</CropBox[0 0 387.36 594]/MediaBox[0 0 387.36 594]/Parent 1 0 R/Rotate 0/Type/Page/VP[<</BBBox[0 0 387.36 594]/Measure<</Bounds[0 0 1 0 1 1 0 1 0 0]/GCS<</EPSG 0/Type/PROJCS/WKT(PROJCS["",GEOGCS["GCS_North_American_1983",DATUM["D_North_American_1983",SPHEROID["GRS_1980",6378137.000,298.25722210]],PRIMEM["Greenwich",0],UNIT["Degree",0.017453292519943295]],PROJECTION["Lambert_Conformal_Conic"],PARAMETER["False_Easting",0.000],PARAMETER["False_Northing",0.000],PARAMETER["Central_Meridian",-73.69316666666671],PARAMETER["Latitude_Of_Origin",40.72530555555556],PARAMETER["Standard_Parallel_1",44.66666666666671],PARAMETER["Standard_Parallel_2",39.33333333333333],UNIT["Inch",0.02540005080010]))>>/GPTS[40.13654277517 -74.01557170405 40.13597685236 -73.21340679682 41.08021243664 -73.20641179920 41.08078674605 -74.02027249101]/LPTS[0 0 1 0 1 1 0 1]/PDU[(MI)(SQMI)(DEG)]/Subtype/GEO/Type/Measure>>/Type/Viewport>>]>>endobj
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Recommendation

It's unnecessary to change the current PDFs. The LPTS positions in the Current PDFs are fine as they stand, because any software that is able to interpret Geospatial PDF already understands how to relate LPTS and GPTS. **The PDFs are fine as-is.** If any change is to be made, I suggest pinning the LPTS at the corners of a rectangle enclosing the plan view, although this is not crucial. It would be a better use of time to set the Bounds to the plan view's polygon.

For external georeferencing data. If control points are to be extracted and made available in another format, such as CSV or georeferenced d-TPP_Metatile.xml, these ".8" points extracted from the PDF's LPTS are not the best choice. Software consuming such external data may be expecting *only* corner coordinates for control points, and might not be able to make good use of these arbitrary ".8" points. **For external data, corner points are ideal.**



00610IL22L.PDF

